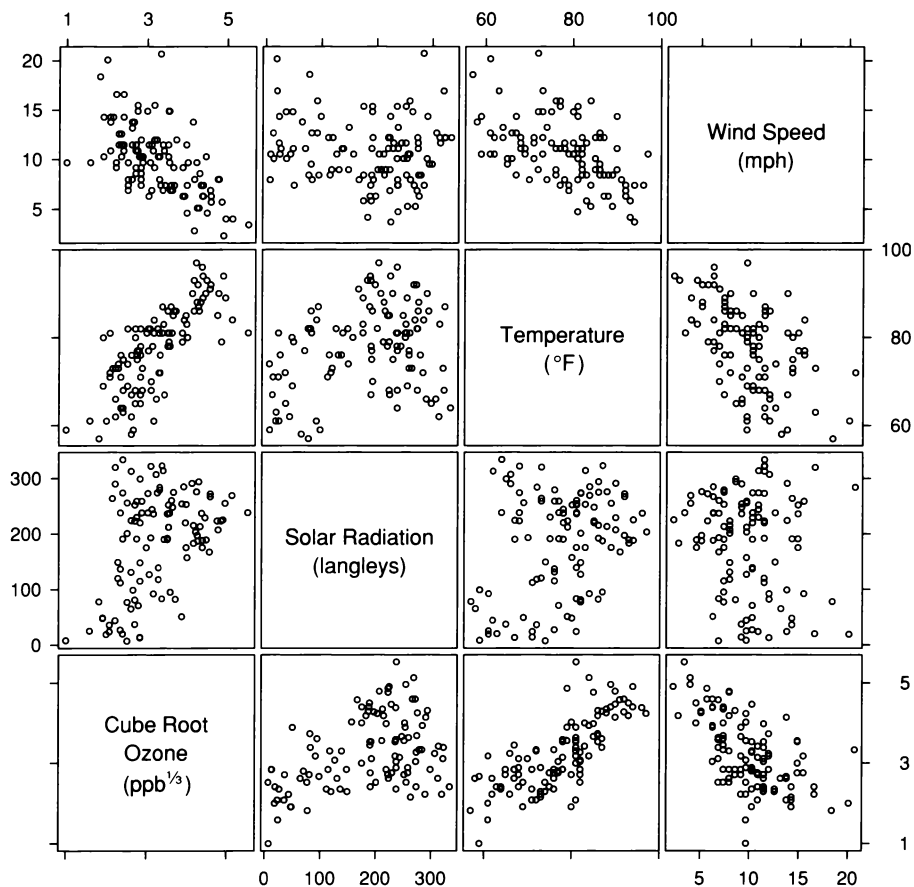




5.1 A scatterplot matrix graphs hypervariate data: measurements of solar radiation, temperature, wind speed, and cube root ozone concentration on 111 days at sites in the New York metropolitan region.

5 *Hypervariate Data*

Figure 5.1 is a scatterplot matrix of measurements of four environmental variables from an air pollution study [13]. The variables, all measured at ground level, are wind speed, temperature, solar radiation, and cube root ozone concentration. Ozone is one of the nasty constituents of photochemical smog, and its concentration is the standard indicator of the severity of such smog. If its level is high, a smog alert is called. Ozone is not emitted directly into the atmosphere. Rather, it is a product of chemical reactions that require solar radiation and emissions of primary pollutants from smoke stacks and automobile tail pipes. When the ventilation of the atmosphere is low, the chemical reactions bring ozone to high levels. Low ventilation tends to occur when wind speeds are low and temperatures are high because on hot, calm days in the summer the atmosphere cannot cleanse itself. The goal in analyzing the environmental data is to determine how ozone depends on the other variables, so cube root concentration is the response and the other variables are factors.

The environmental data in Figure 5.1 lie in a four-dimensional space. In Chapter 2 we visualized univariate data, which lie on a line; in Chapter 3 it was bivariate data, which lie in the plane; and in Chapter 4 it was trivariate data, which lie in three-space. The data in these cases lie in spaces with physical reality. We exploit this reality to visualize the data. The environmental data are hypervariate; they have too many variables for visual reality. We must peer cognitively, not perceptually, into hypervariate space by looks at subspaces of dimension three or fewer. In classical statistics, the word “multivariate” means two or more variables, and was coined because probabilistic methods have a natural breakpoint between one dimension and two. For visualization, we need the term “hypervariate” to acknowledge the visual breakpoint between three and four dimensions.

The measurements in Figure 5.1 were made on 111 days from May to September of 1973 at sites in the New York City metropolitan region; there is one measurement of each variable on each day. Solar radiation is the amount from 0800 to 1200 in the frequency band 4000–7700Å, and was measured in Central Park, New York City. Wind speed is the average of values at 0700 and 1000, and was measured at LaGuardia Airport, which is about 7 km from Central Park. Temperature is the daily maximum, and was also measured at LaGuardia. Ozone is the cube root of the average of hourly values from 1300 to 1500, and was measured at Roosevelt Island, which is about 2 km from Central Park and 5 km from LaGuardia. The cube root transformation symmetrizes the distribution of the concentrations; on the original scale, they are significantly skewed toward large values.

To streamline the discussion we will use the following notation: O_3 = cube root ozone concentration, R = radiation, T = temperature, and W = wind speed.

5.1 Scatterplot Matrices

An award should be given for the invention of the *scatterplot matrix*, but the inventor is unknown — an anonymous donor to the world's collection of visualization tools. Early drafts of *Graphical Methods for Data Analysis* [16] contain the first written discussion of the idea, but it was in use before that. The inventor may not have fully appreciated the significance of the method or may have thought the idea too trivial to bring it forward, but its simple, elegant solution to a difficult problem is one of the best visualization ideas around.

As with other multi-panel displays in this book, we will refer to a panel of the scatterplot matrix by column and row number; the left column is column one and the bottom row is row one. For example, in Figure 5.1, the upper left panel is (1,4) and the lower right is (4,1).

Visual Linking

One way to visualize hypervariate data is to graph each pair of variables by a scatterplot. But just making the scatterplots without any

coordination often results in a confusing collection of graphs that are hard to visually integrate.

The important idea of the scatterplot matrix is to arrange the graphs in a matrix with shared scales. As a result, we can visually *link* features on one scatterplot with features on another, which greatly increases the power of the visualization. Along each row or column of the matrix, one variable is graphed against all others with the scales for that one variable lined up along the horizontal or the vertical. By scanning a row or column we can link effects on different scatterplots.

Visual linking in Figure 5.1 allows us to detect important effects in the environmental data. Panel (2,1), a scatterplot of O_3 against R , has an upper envelope in the form of an inverted “V”. There are two interesting properties. First, the highest values of O_3 occur when R is between 200 and 300 langleys. Second, for the very highest values of R , O_3 stays at low levels. Visual linking provides an explanation of both of these properties. First, we can focus on the points with the highest O_3 values in the (2,1) panel, and then scan to the right to panels (3,1) and (4,1). The linking shows that the high O_3 results when T is high and W is low; these are days with low ventilation. Next, we can focus on the points in the (2,1) panel with the highest values of radiation and scan up to the (2,3) and (2,4) panels. We see that for the very highest values of R , T tends not to be high and W tends not to be low, so O_3 does not rise to high levels.

Panels (3,1) and (4,1) of Figure 5.1 show there is a strong association between T and O_3 and between W and O_3 . The reason, as stated earlier, is that W and T are both indicators of ventilation. But the (4,3) panel shows that W and T are related and thus are measuring ventilation, to some extent, in the same way.

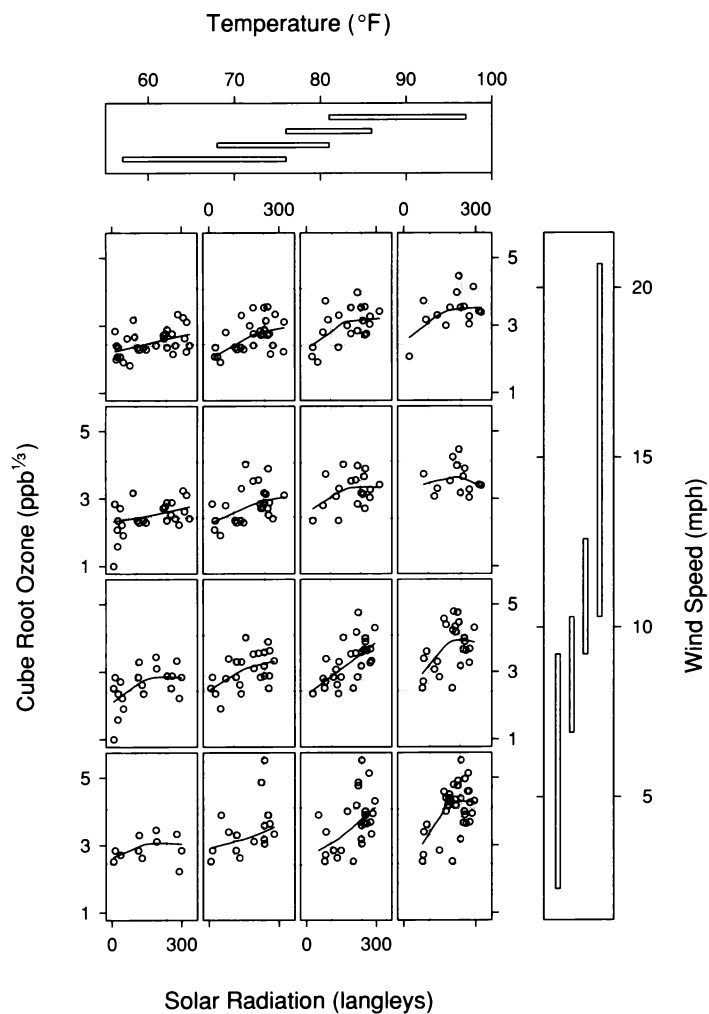
Visual linking is the reason, despite the redundancy, for including both the upper and lower triangles in the scatterplot matrix; the upper left triangle has all pairs of scatterplots, and so does the lower right triangle. Suppose that only the upper left triangle were present in Figure 5.1. To see temperature against everything else, we would have to scan the first two graphs in row three, turn the corner in the (3,3) panel, and go up to row four to see the remaining scatterplot. The three temperature scales would not be lined up, which would interfere with the linking.

5.2 Coplots of Data

The scatterplot matrix of the environmental data shows a strong association between O_3 and T and between O_3 and W . It seems likely that each of these factors explains variation in the data not explained by the others. There is less evidence that R explains variation in O_3 given T and W . For example, we saw that the upper envelope on the scatterplot of O_3 against R was explainable by the behavior of T and W . But the photochemistry that results in O_3 requires radiation, and the laws of chemistry must be obeyed, so we know in principle that radiation is a causal factor. But this does not mean that for the environmental data, the solar radiation measurements are good predictors of O_3 . Perhaps only a minimum amount of radiation is needed, and beyond that, production is not sensitive to the amount. Or it may be that the Central Park measurements do a poor job of characterizing the amount of radiation that actually falls on the air mass.

We need a way to see the dependence of O_3 on radiation given temperature and wind speed. A hypervariate coplot, an extension of the coplot display method of Chapter 4 to three or more factors, can provide such a conditional visualization. Figure 5.2 shows such a coplot. O_3 is graphed against R given W and T . Each dependence panel is a scatterplot of O_3 against R for days with both W and T in specified intervals. The given panels, which are at the top and right of the graph, show the intervals. The four scatterplots in any row of the 4×4 array of dependence panels have the same conditioning interval of values of W ; as the rows go from bottom to top, the intervals of W increase. Similarly, the four scatterplots in any column have the same conditioning interval of values of T ; as the columns go from left to right, the intervals of T increase.

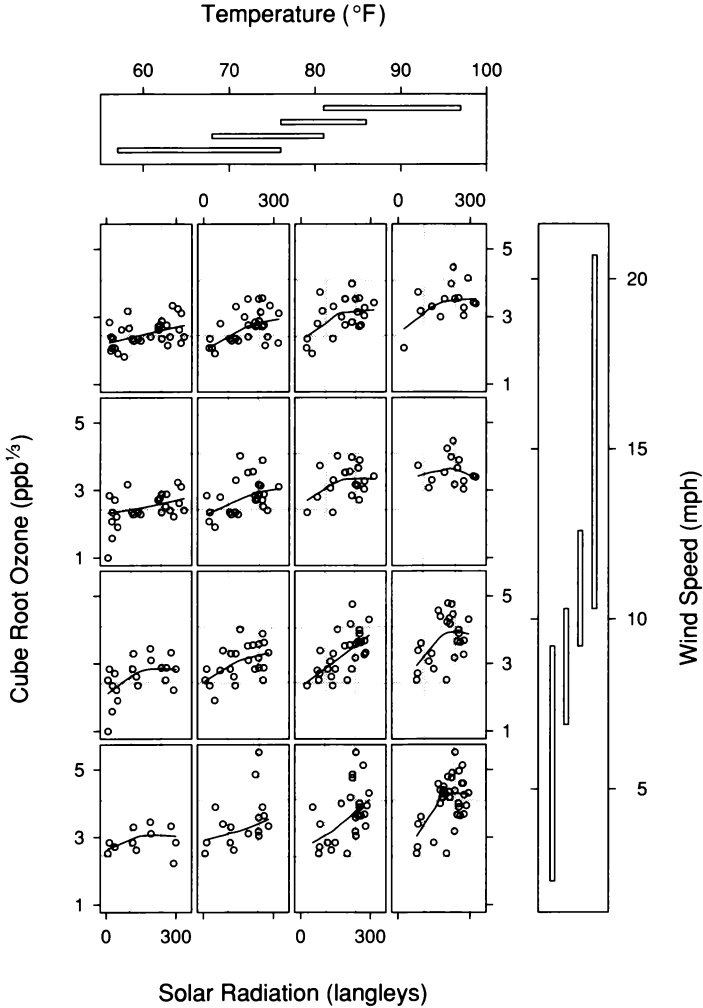
Each panel in Figure 5.2 has a loess curve to help us visualize the dependence pattern. The fitting of the curves uses bisquare; as with coplots for two factors, it is prudent to use robust fitting for initial looks at the data to protect against distortion that might occur if there happen to be outliers.



5.2 A coplot graphs O_3 against R given T and W . The robust loess curves, whose parameters are $\lambda = \alpha = 1$, are banked by an aspect ratio of 2. Banking to 45° results in an aspect ratio that is too large.

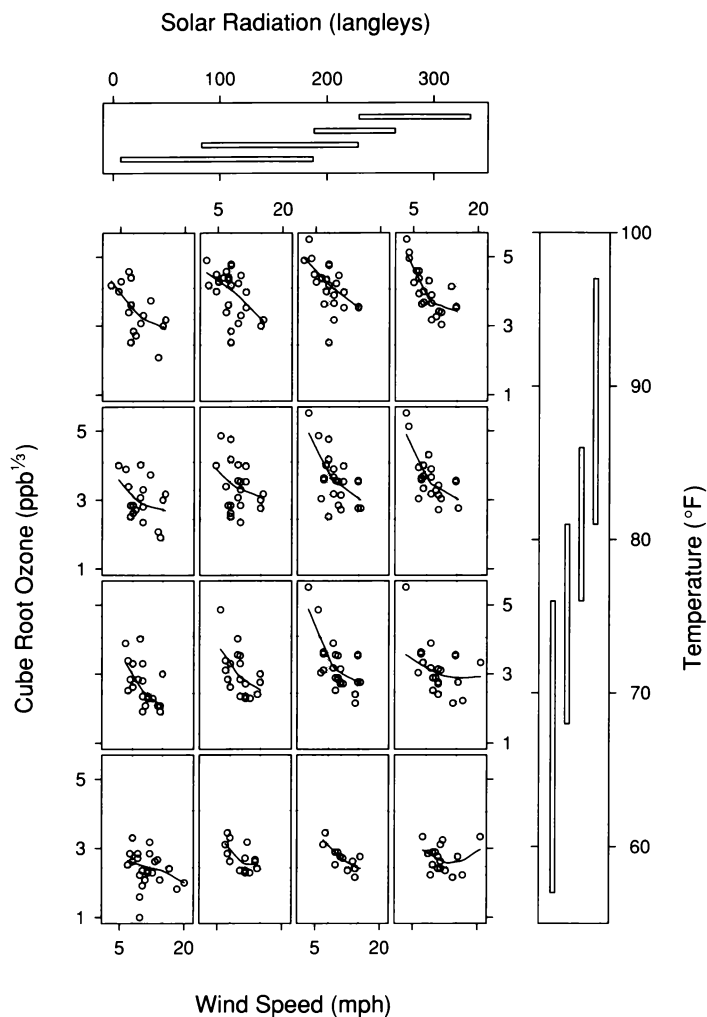
The coplot of Figure 5.2 is repeated in Figure 5.3 to enhance the following discussion. The conditioning intervals of Figure 5.3 have been chosen using the equal-count method with a target fraction of overlap equal to $1/2$. Thus, the number of values of the conditioning factor in each interval is roughly constant. The interval counts for T are 46, 51, 51, and 51. For W , they are 49, 50, 47, and 53. However, because the points on each dependence panel arise from satisfying two conditionings, one for T and one for W , the number of points on the dependence panels varies from 11 to 34. Consider the (1,1) panel, a scatterplot of O_3 against R for observations that satisfy two requirements: the values of T must be in the lowest temperature interval and the values of W must be in the lowest wind speed interval. There are 49 days with W in the wind interval and 46 days with T in the temperature interval. If it happened that all days with T in the temperature interval also had W in the wind interval, there would be 46 points on the dependence panel. But if it happened that no days with T in the temperature interval had W in the wind interval, there would be no points on the dependence panel. Any number between these two extremes is possible, and the actual value depends on how W and T vary with one another. Since low temperature and low wind speed tend not to occur together, the number of points on panel (1,1) is only 11. But on the (4,1) panel, where T is high and W is low, there are 34 points since high temperatures are associated with low wind speeds.

Figure 5.3 shows rather convincingly that radiation explains variation in the O_3 measurements beyond that explained by wind speed and temperature. As the theory of photochemistry predicts, O_3 increases with R . The patterns of dependence are generally concave. Thus, for the most part, the marginal effect of R diminishes as R increases. The shapes of the patterns vary. For each fixed value of W , the overall change as a function of R increases as T increases. This means there is an interaction between R and T — as the ventilation drops, the effect of R is greater. There is a suggestion of a similar interaction between W and R — a greater overall change as a function of R as W decreases, possibly for the two largest values of T — but there is too much noise in the data to be certain.



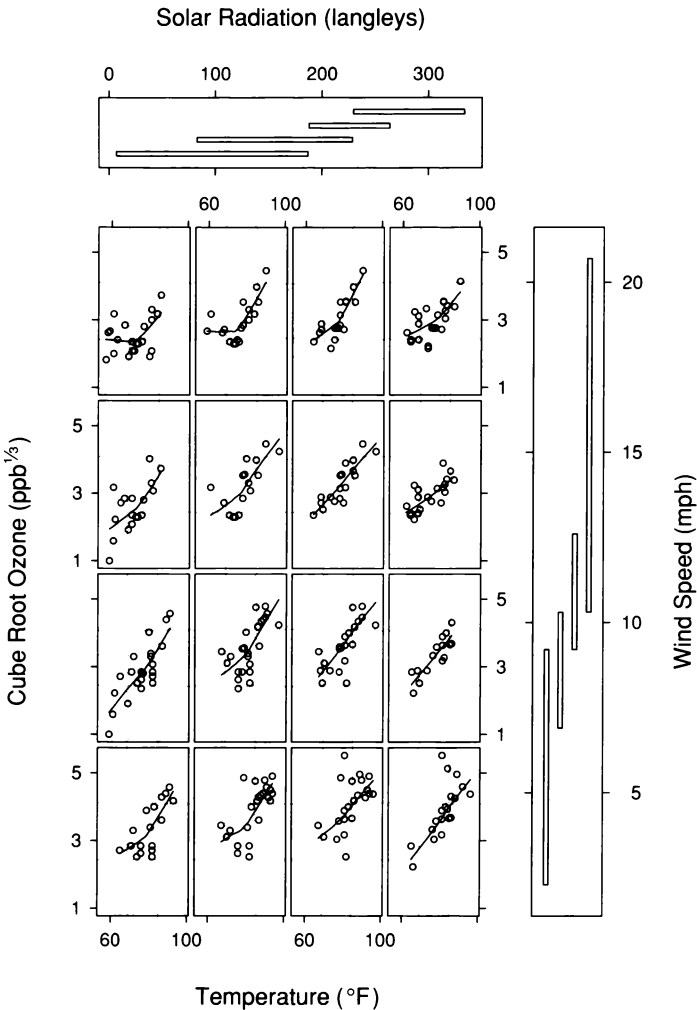
5.3 The coplot of O_3 against R given T and W is displayed again.

For three factors, there are three coplots; each factor appears once on the horizontal scale. Figures 5.4 and 5.5 are the remaining two coplots for the environmental data. Figure 5.4 shows, as expected, that as W increases, O_3 decreases. The patterns are mostly convex; the marginal effect of W diminishes as W increases. Careful examination of each column of panels does not clearly establish any interaction between W and T . The assessment must be done with care because the distribution of values of W on each panel shifts toward lower values as the conditioning



5.4 O_3 is graphed against W given R and T . The robust loess curves, whose parameters are $\lambda = \alpha = 1$, are banded by an aspect ratio of 2.

value of T increases for fixed R . But an examination of portions of curves that overlap in W does not suggest much change in shape. Also, as in the previous coplot, there is no clear indication of an interaction between R and W . Figure 5.5 shows that O_3 increases markedly with increasing T . Unlike for R and W , there is no reduction in the marginal effect for large values. The interaction between R and T is again apparent. But the change in shape of the curves with W for fixed R appears to be due more to a changing distribution of values of T on the panels than to a meaningful interaction.



5.5 O_3 is graphed against T given R and W . The robust loess curves, whose parameters are $\lambda = \alpha = 1$, are banked by an aspect ratio of 2.

5.3 *Coplots of Hypervariate Surfaces and Cropping*

Fitting is essential to visualizing hypervariate data. The structure of data in many dimensions can be exceedingly complex. The visualization of a fit to hypervariate data, by reducing the amount of noise, can often lead to more insight. The fit is a hypervariate surface, a function of three or more variables. As with bivariate and trivariate data, our fitting tools are loess and parametric fitting by least-squares. And each tool can employ bisquare iterations to produce robust estimates when outliers or other forms of leptokurtosis are present.

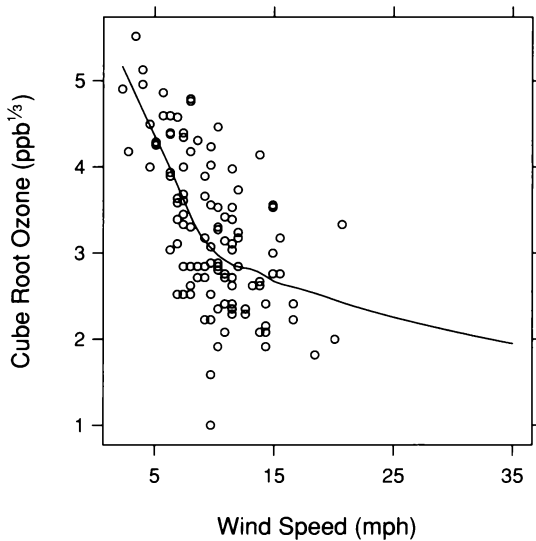
Loess fitting provides a satisfactory fit to the environmental data. The fit is a hypervariate surface, $\hat{g}(R, W, T)$, a function of radiation, wind speed, and temperature. Extensive experimentation led to this fit, whose parameters are $\alpha = 1$ and $\lambda = 2$. Bisquare was not used since, as we shall see shortly, the residual distribution is well approximated by the normal. In Chapter 4, the ethanol data were fitted by a loess surface that was conditionally linear in compression ratio. For the environmental data, the loess fit was taken to be conditionally quadratic in W and R ; this means that given T , the fit as a function of W and R is a quadratic polynomial whose coefficients change as the conditioning value of T changes. The coplots of the data in Figures 5.3 to 5.5 led to this specification; the coplots showed that the patterns of dependence on W and R are mildly curved, but the dependence on T is both greater and more complex.

Cropping

There is an insidious problem in visualizing a fit to hypervariate data — restricting our look at the fit to regions of the factor space that have sufficient data. The problem occurs for bivariate data and trivariate data as well, but it is easier to assess in these cases and solutions are simpler. For hypervariate data, the problem is treacherous because it is easy to view a hypervariate fit and not appreciate that certain portions of the fit are evaluations over regions with little or no data.

Consider first, the bivariate case. Suppose the environmental data had just one factor, wind speed. Figure 5.6 shows a loess fit to the data.

Something leaps out at us that most would regard as needing explicit justification; the curve has been extrapolated well beyond the maximum measured value of W . It would take a large leap of faith or substantial knowledge of the system under study to believe that the fit can be extended so far beyond the measurements of W . In Chapter 3, fits to bivariate data were routinely computed over a range that extended from the minimum value of the factor to the maximum. And in all cases, there were no unduly large observation gaps within the range of the data. Thus, visualized fits were supported by the data. If gaps had occurred, we would have *cropped* the visualization, showing the fits at less than the full range of the data.

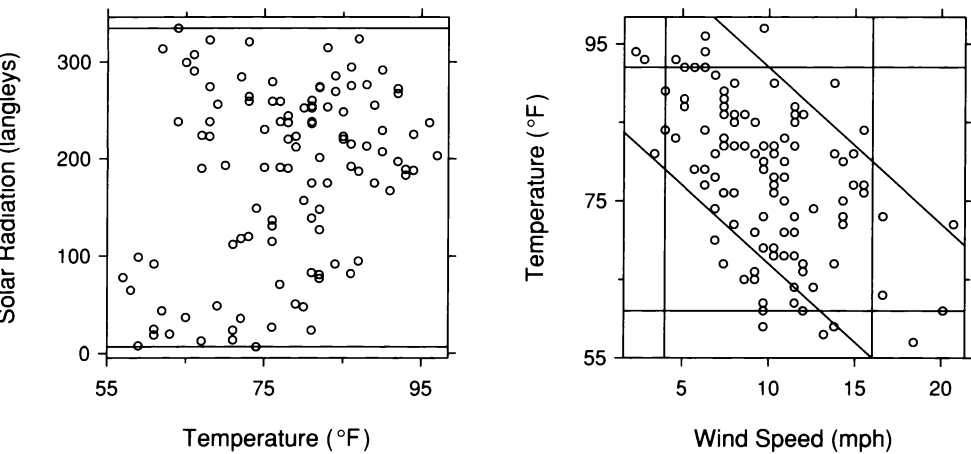


5.6 A portion of the loess fit is evaluated at values of the factor where there are no measurements. The parameters of the fit are $\alpha = 2/3$ and $\lambda = 1$.

For trivariate data, a scatterplot of the two factors shows clearly where the measurements lie in the factor space. For the ethanol data and the perception data, the fits were evaluated over the data rectangle, the rectangle which just encloses the data. In these examples, the data populated the data rectangles in all regions. But for the rubber data, the galaxy data, and the soil data, the data rectangles were cropped; in each of these examples, the fit was evaluated over a rectangle that lay inside the data rectangle. Note that cropping does not imply that data are discarded outside of the cropped region, but simply that we do not study the fit outside of the region, although dropping isolated data can also be prudent, as the rubber data illustrated.

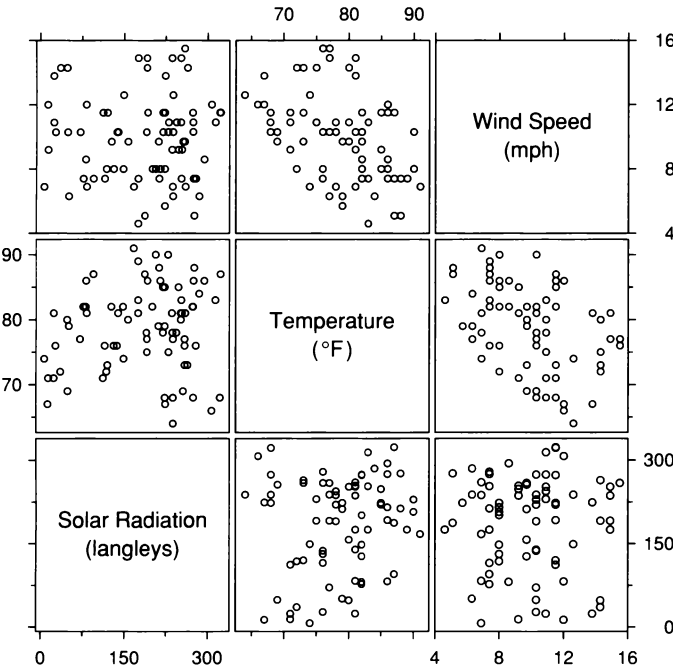
For hypervariate data, it is a greater challenge to determine when and how to crop because the factors lie in spaces of dimension three or higher, which are harder to visualize.

For the loess O_3 fit, $\hat{g}(R, W, T)$, it is at least clear that cropping is needed. The scatterplot matrix of the data in Figure 5.1 shows that the three-dimensional box that just contains the values of the three factors has regions that are far from the measurements. In part, the problem is caused by the correlation between W and T . We could crop with a box that lies inside the data box and whose sides are parallel to those of the data box, but this parallel cropping would also eliminate regions with sufficient data. Instead, we will use the cropping shown in Figure 5.7. The two scatterplots have four pairs of parallel lines that are projections of parallel planes in the three-dimensional space of the factors. The cropped region is the intersection of the four regions between the four pairs of planes. W is cropped by moving toward the center of the data from the extremes, and T is cropped in a similar fashion. W and T are also simultaneously cropped as shown by the two oblique lines. R is not cropped. The cross section of the cropped region perpendicular to the R axis is the six-sided polygon that is formed in the interior of the right panel of Figure 5.7 by the six lines. This cropping was done visually, just as a photographer visually crops a photo for better effect.



5.7 The four pairs of parallel lines indicate the cropped evaluation region for the loess fit to the environmental data.

Figure 5.8 is a scatterplot matrix of the factor measurements that lie inside the cropped region. No unduly large gaps appear; the (1,2) panel does show some small regions that are empty, but we can tolerate this. Of course, the scatterplots of the scatterplot matrix are projections, and thus could mask gaps in the three-dimensional cropped region, but rotation of the 3-D point cloud, a direct manipulation method discussed in Chapter 4, showed that the cropped region is populated throughout with data.



5.8 The scatterplot matrix displays those measurements of the environmental factors that lie inside the cropped region.

Displaying Hypervariate Surfaces

The strategy for displaying a fitted hypervariate surface, $\hat{g}$, is to use conditioning. Suppose u is one of the factors. We can assign values to all other factors, and graph $\hat{g}$ against u given these assigned, or conditioning, values. The result is a curve that can be displayed by a bivariate graph of $\hat{g}$ against u . We must vary the conditioning values in a systematic way to visualize the surface. This can be done using the format of the hypervariate coplots of the environmental data in

Figures 5.3 to 5.5, except that the conditioning intervals are replaced by single conditioning values. The result is a hypervariate coplot of a hypervariate surface, an extension of the coplot method for displaying surfaces that was introduced in Chapter 4.

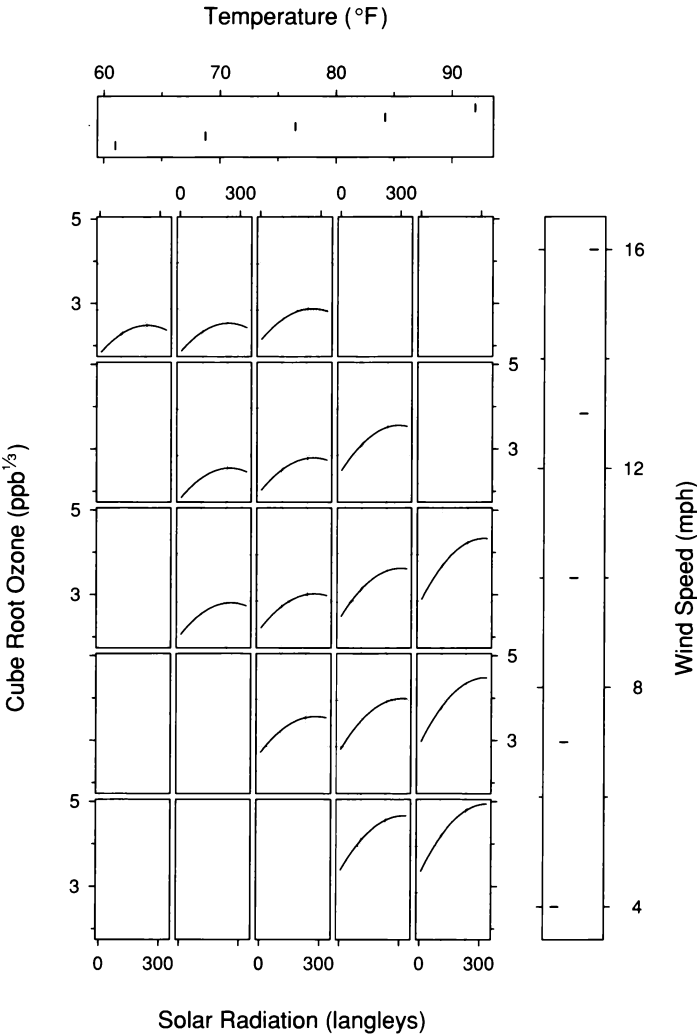
We can also graph against two factors given the others. Suppose v is a second factor. We can condition on all factors except u and v , and graph $\hat{g}$ as a function of u and v by one of the factor-plane methods of Chapter 4, for example, by a level plot. Again, the conditioning values must be systematically varied to produce a hypervariate factor-plane display [74].

Figure 5.9 is a hypervariate coplot that displays the loess O_3 surface, $\hat{g}(R, W, T)$, over the cropped region displayed in Figure 5.7. $\hat{g}$ is graphed against R conditional on W and T . For each panel with a curve, T and W have been set to specific values and the curve on the panel is a graph of the fit against R . The fixed value of T for a panel is shown above the column to which the panel belongs. The fixed value of W is shown to the right of the row to which the panel belongs. For example, for the (1,5) panel, T is 61°F and W is 16 mph. The five conditioning values of W are equally spaced from 4 mph to 16 mph, the two cropping values for W . Similarly, the five conditioning values of T range from 61°F to 92°F, the two cropping values for T . The curves have been evaluated at 50 equally spaced values of R from 7 langleys, the minimum observed value of R in the data, to 334 langleys, the maximum observed value. The minimum and the maximum are the boundaries of the cropped region in the R direction. On some panels, no curves occur because the combinations of T and W put them outside the cropped region.

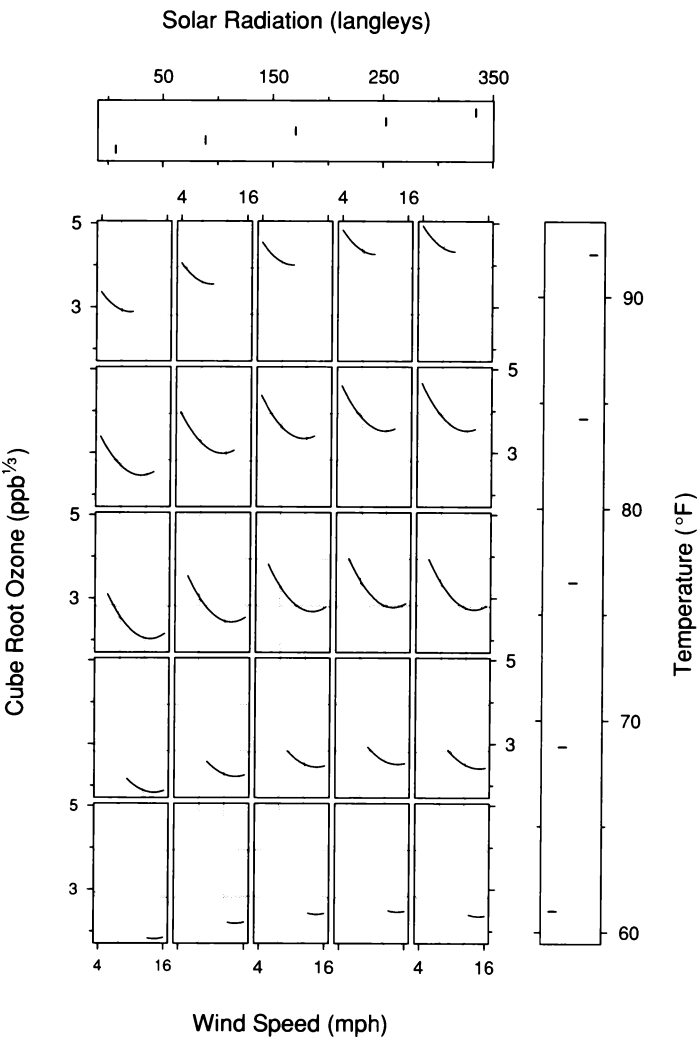
Figure 5.10 is a coplot of the O_3 fit against W given R and T . Figure 5.11 is a coplot of the O_3 fit against T given R and W . For these two displays, the cropping results in curves on all panels, but with varying evaluation intervals.

The three cropped coplots show general patterns that are similar to those revealed by the coplots of the data, but now the patterns are conveyed more incisively since the noise of the data has been removed. Figure 5.9 shows a nonlinear dependence of O_3 on R with a decreasing marginal effect of R . There is an interaction between R and T ; as T increases for fixed W , the overall change in O_3 increases. There does not

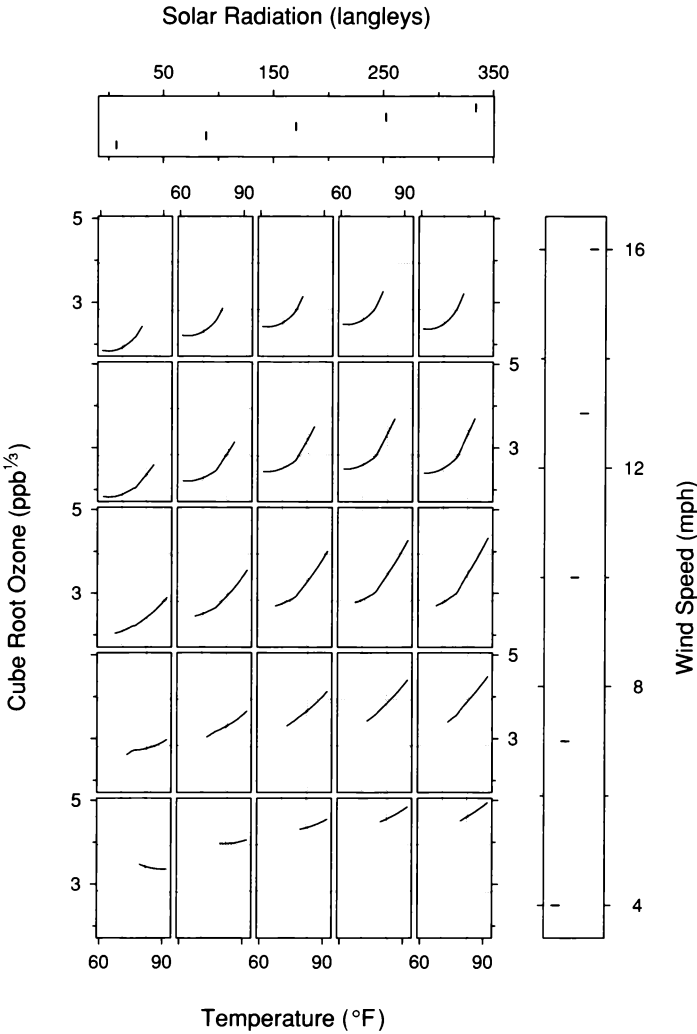
appear to be an interaction between R and W . Figure 5.10 reveals a nonlinear dependence of O_3 on W with a decreasing marginal effect of W . There are no convincing interactions. The change in shape of the curves along each column appears to be due to the change in the evaluation interval; where there is overlap in evaluation, the curves appear to have the same shape. Figure 5.11 reveals both a strong dependence of O_3 on T and the interaction between T and R that we saw in Figure 5.9.



5.9 A coplot graphs fitted O_3 against R given T and W . The curves are banded by an aspect ratio of 2.



5.10 A coplot graphs fitted O_3 against W given R and T . The curves are banked by an aspect ratio of 2.

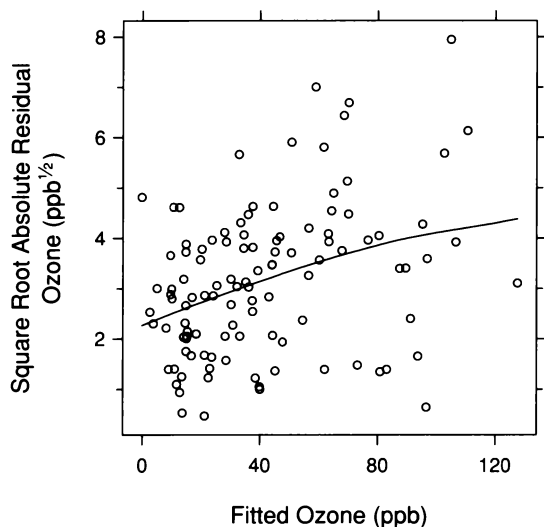


5.11 A coplot graphs fitted O₃ against *T* given *R* and *W*. The curves are banked by an aspect ratio of 2.

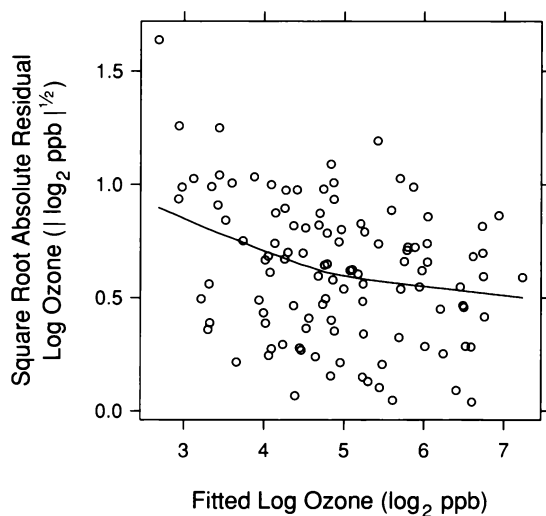
Graphing Residuals

The loess O_3 surface appears to fit the data because six displays (not shown here) did not reveal lack of fit — three scatterplots of the residuals against the factors and three coplots of the residuals.

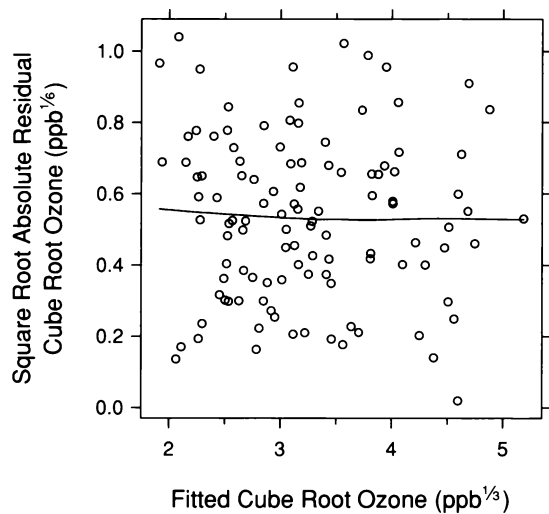
The analysis of the environmental data began with a transformation; taking cube roots of the ozone concentrations symmetrized their distribution. Good fortune was with us because the cube root transformation also cures monotone spread. Figure 5.12 is an s-l plot for a loess fit to concentration without transformation; the parameters are the same as those for the fit to the cube roots. There is monotone spread, an increase in the spread with the fitted values. Logs do not cure the monotone spread. Figure 5.13 is an s-l plot for a loess fit to log concentration with the same parameters as the other fits. The spread decreases as the fitted values increase. Figure 5.14 is an s-l plot for the fit to the cube roots. The spread of the residuals is uniform. The cube root has struck the proper balance.



5.12 An s-l plot checks for monotone spread for the loess fit to concentration. The parameters of the loess curve on the graph are $\alpha = 1$ and $\lambda = 1$.



5.13 An s-l plot checks for monotone spread for the loess fit to log concentration. The parameters of the loess curve on the graph are $\alpha = 1$ and $\lambda = 1$.



5.14 An s-l plot checks for monotone spread for the loess fit to cube root concentration. The parameters of the loess curve on the graph are $\alpha = 1$ and $\lambda = 1$.